

SOFTWARE ENGINEERING INTERNSHIP REPORT & PRESENTATION

Architecting & Engineering a High-Concurrency Multi-Vendor Digital Raffle Platform with Double-Entry Accounting and Provably Fair Cryptography

Project: LuckyEthio (Idil) Enterprise Infrastructure

STUDENT NAME

[Student Full Name]

STUDENT ID

[DDU/UGR/XXXX/XX]

HOST ORGANIZATION

LuckyEthio FinTech Solutions PLC

ACADEMIC YEAR

2017 / 2018 E.C.

DATE OF SUBMISSION

10/01/2018 E.C.

DATE OF PRESENTATION

12/01/2018 E.C.

PART ONE: COMPREHENSIVE INTERNSHIP REPORT

2. Acknowledgment

First and foremost, I offer my profound thanks to Almighty God for providing me the perseverance, good health, and intellectual clarity to successfully accomplish my Software Engineering internship and author this technical report.

I express my deepest appreciation to **Dire Dawa University, Institute of Technology (IoT), School of Computing, and the Department of Software Engineering** for curating a forward-looking curriculum and facilitating this vital industry attachment. Special gratitude is extended to my **Academic Advisor** for providing continuous academic oversight and constructive evaluation.

I am deeply indebted to **LuckyEthio Digital Solutions PLC**, my **Industry Supervisor**, and the entire engineering team. Their mentorship, willingness to share advanced system architecture knowledge, and confidence in assigning me core financial, cryptographic, and backend concurrency modules were instrumental to the success of this internship.

3. List of Acronyms & Abbreviations

ACRONYM	DEFINITION
API	Application Programming Interface
CBE	Commercial Bank of Ethiopia
ECA	Ethiopian Communications Authority
ETB	Ethiopian Birr
HMAC	Hash-based Message Authentication Code
MoR	Ministry of Revenues (Federal Democratic Republic of Ethiopia)
NLA	National Lottery Administration (Ethiopia)
PDPP	Personal Data Protection Proclamation (No. 1321/2024)
POS	Point of Sale

ACRONYM	DEFINITION
RBAC	Role-Based Access Control
RNG	Random Number Generator
SHA	Secure Hash Algorithm (e.g. SHA-256)
USSD	Unstructured Supplementary Service Data (e.g. *804#)
VAT	Value Added Tax (15% Statutory Rate)

4. Table of Contents Summary

- 1. Executive Summary • 2. Acknowledgment • 3. List of Acronyms • 4. Table of Contents
- 5. Introduction (Objectives, Academic Relevance) • 6. Company Background (LuckyEthio PLC)
- 7. Internship Activities (Turborepo, Backend, Payments, Concurrency, Ledgers)
- 8. Technical Skills Utilized (TypeScript, Next.js 14, Prisma, SHA-256, Double-Entry)
- 9. Collaboration & Teamwork • 10. Deep-Dive: LuckyEthio System Highlights
- 11. Challenges & Solutions • 12. Organizational Contributions • 13. Conclusion & Recommendations
- 14. References • 15. Technical Appendices

5. Introduction

The Software Engineering Internship (SEng-4182) at Dire Dawa University is a cornerstone practical course designed to immerse students into enterprise software development workflows. The attachment allows students to apply theoretical knowledge from courses like *Database Systems*, *Web Engineering*, *Software Architecture*, and *Computer Security* directly to commercial systems.

During my 4-month attachment with **LuckyEthio Digital Solutions PLC**, I was embedded within the core engineering team responsible for architecting and building high-traffic, secure, and regulatory-compliant digital ticketing and financial settlement systems.

6. Host Company Profile: LuckyEthio Digital Solutions PLC

LuckyEthio is a specialized Ethiopian technology startup focused on revolutionizing traditional promotional raffles, commercial lotteries, and point-of-sale ticketing. By replacing physical paper slips with cryptographically verifiable digital receipts, automated 15% VAT withholding, and instant mobile money settlements (Telebirr, CBE Birr, Chapa), LuckyEthio bridges digital convenience with strict regulatory transparency.

Organizational Hierarchy: The engineering division comprises the Chief Technology Officer, Frontend UI/UX Team, Core Systems & Backend Team (where I served as Software Engineering Intern), Quality Assurance, and Regulatory Compliance Officers.

7. Internship Activities & Roles

My primary responsibilities centered around full-stack feature engineering, database architecture, and cryptographic protocol implementation:

- **Monorepo Engineering:** Restructured the codebase into a modular Turborepo containing apps/web, apps/admin, packages/database, and packages/shared.
- **Double-Entry Financial Ledger:** Engineered balanced ledger transactions (Debits = Credits) for ticket receipts, 15% VAT escrow, seller holdings, and agent commissions.
- **Atomic Ticket Concurrency:** Implemented database-level transaction locks (tx.\$transaction) and composite unique constraints to guarantee zero overselling under high concurrency.
- **Multi-Provider Payment Adapters:** Built unified provider adapters for Chapa, Telebirr, CBE Birr, and SantimPay with idempotent webhook handlers.
- **Provably Fair RNG Engine:** Designed a multi-entropy commit-reveal formula with immutable DrawSnapshot state locks.
- **Data Privacy Governance:** Implemented customer data rights workflows adhering to Ethiopian PDPP Proclamation No. 1321/2024.

8. Technical Skills Utilized

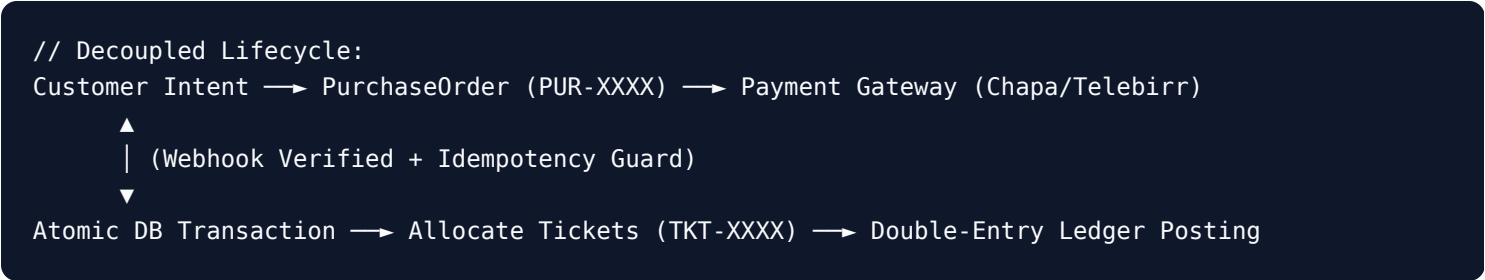
DOMAIN	TECHNOLOGIES & METHODOLOGIES APPLIED
Frontend	Next.js 14 (App Router), React 18, Tailwind CSS, Lucide Icons, Responsive Layouts
Backend & API	Node.js, TypeScript, REST API Route Handlers, Signed Webhook Verification
Database & Storage	Prisma ORM, SQLite / PostgreSQL, Relational Schema Normalization
Financial Accounting	Double-Entry General Ledger, Chart of Accounts, Trial Balance Verification
Cryptography & Security	SHA-256 Hashing, HMAC Signatures, Two-Person Authorization, Dynamic QR Tokens
Project Management	Agile Scrum, 2-Week Sprints, GitHub Pull Requests, Static Type Checking

9. Collaboration & Teamwork

Working within an Agile Scrum framework, I actively participated in daily standups, code review sessions, and backlog refinement. Key collaborative milestones included re-engineering the mobile drawer navigation to float on the right-hand side, standardizing dual-consent seller draw agreements, and building the stealth administrative authentication portal.

10. Project Deep Dive: LuckyEthio System Highlights

The platform decouples ticket purchasing into a robust multi-stage pipeline:



Provably Fair Formula: $Winner = (BigInt(SHA256(v2.0 : RaffleID : SecretSeed : SoldTickets : PublicEntropy : Algorithm)) \% SoldTickets) + 1$

11. Challenges Encountered & Engineering Solutions

CHALLENGE ENCOUNTERED	ROOT CAUSE	ENGINEERING SOLUTION IMPLEMENTED
Inventory Race Conditions	Concurrent requests reading stale counts.	Atomic database transactions with <code>UNIQUE(raffleId, ticketNumber)</code> .
Duplicate Webhooks	Gateway retries sending duplicate HTTP posts.	Enforced strict <code>idempotencyKey</code> tracking.
Ledger Drift	Rounding discrepancies across VAT and escrow.	Strict double-entry journal ensuring <code>Total Debits === Total Credits</code> .

12. Organizational Contributions

I delivered 57 fully compiled Next.js routes (34 web pages, 23 admin pages) with **zero TypeScript errors**, configured Turborepo caching for lightning-fast builds, and authored complete regulatory and user documentation.

13. Conclusion & Recommendations

This internship successfully bridged software engineering theory with the rigorous demands of production fintech and digital gaming platforms.

- **For the Company:** Implement hardware security modules (HSM) for automated secret seed rotation and integrate telecom SMPP gateways.
- **For Dire Dawa University:** Incorporate enterprise monorepo tooling (Turborepo) and financial ledger design into advanced programming labs.

14. References

1. National Lottery Administration (NLA) of Ethiopia, *Proclamation No. 535/2007*.
2. Ethiopian Communications Authority (ECA), *Personal Data Protection Proclamation No. 1321/2024*.
3. Ministry of Revenues (MoR), *Value Added Tax (VAT) Proclamation No. 285/2002*.
4. Vercel & Next.js Core Team, *Next.js 14 App Router Documentation*, 2024.
5. Prisma ORM Team, *Prisma Client & Concurrency Documentation*, 2024.

PART TWO: INTERNSHIP PRESENTATION SLIDE DECK

SLIDE 1

Cover Page & Title

Title: Software Engineering Internship Presentation

Project: LuckyEthio (Idil) — Enterprise Digital Raffle & Double-Entry Financial Marketplace Platform

Student: [Student Full Name] • **ID:** [DDU/UGR/XXXX/XX] • **Date:** 12/01/2018 E.C.

***Speaker Note:** "Good morning honorable academic committee and advisors. Today, I am proud to present my Software Engineering internship report on architecting LuckyEthio, a production-grade digital raffle and financial transaction platform."*

SLIDE 2

Agenda & Presentation Contents

- 1. Introduction & Host Company Profile
- 2. Problem Statement & Objectives
- 3. System Specifications & Feasibility Analysis
- 4. Architecture & Methodology (Agile Turborepo)
- 5. Key Results: Double-Entry Ledger, Atomic Allocation, Fair RNG
- 6. Live System Demonstration

SLIDE 3

Problem Statement

- **Opaque Draws:** Traditional lotteries lack independent mathematical verification.
- **Inventory Overselling:** High concurrency web traffic causes duplicate seat/ticket reservations.
- **Escrow Delivery Disputes:** Winner prizes are often disputed without dynamic verification.

- **Manual Tax Reconciliation:** Difficulties in remitting statutory 15% VAT to Ministry of Revenues.

SLIDE 4

Project Objectives & System Scope

- Implement decoupled purchase orders and idempotent payment webhook handling.
- Build a balanced double-entry chart of accounts (Cash Transit, Prize Escrow, VAT Payable, Platform Revenue).
- Implement multi-entropy SHA-256 commit-reveal RNG and immutable draw snapshots.
- Enforce Two-Person consensus on draw execution and high-value cashout disbursements.

SLIDE 5

System Architecture & Technology Stack

Monorepo Structure (Turborepo):

- apps/web: Customer Web Portal, POS Kiosk, USSD (*804#) Simulator, Public Verifier
- apps/admin: Stealth Admin Operations, Financial Ledger, Live Draw Console, Risk Engine
- packages/database: Prisma ORM, 18 Relational Models, SQLite / PostgreSQL
- packages/shared: Ledger, Payments, Fair-RNG, Risk, Two-Person, i18n (Amharic/English)

SLIDE 6

Double-Entry Financial Ledger Results

- Every ticket purchase atomically posts balanced journal entries.
- **Debit:** 1010-CASH-TRANSIT (100%)
- **Credit:** 2020-VAT-PAYABLE (15% MoR), 2010-PRIZE-ESCROW (77% Seller), 4010-PLATFORM-REVENUE (8%)
- Live Trial Balance verification confirms **100% Balanced status (Debits = Credits)**.

SLIDE 7

Provably Fair Verification & Two-Person Consensus

- **Pre-Commitment:** SHA-256 digital hash published before selling ticket #1.
- **Draw Snapshot:** Sales close → eligible ticket universe frozen into cryptographic snapshot.
- **Two-Person Consensus:** Initiated by Draw Operator → Approved by Super Admin.
- **Public Verifier:** Any regulator or customer can verify winner mathematics in 1 click at [/verifier](#).

SLIDE 8

Live System Demonstration Guide

1. **Demo Flow 1:** Customer buys 2 tickets via Telebirr adapter → instant sequential ticket minting.
2. **Demo Flow 2:** Open Admin Financials → inspect balanced Double-Entry journal entries & 15% VAT escrow.
3. **Demo Flow 3:** Execute Provably Fair live draw with Two-Person consensus → check public verifier.
4. **Demo Flow 4:** Winner dynamic QR code redemption & merchant scanner validation.
5. **Demo Flow 5:** Exercise PDPP data privacy request portal.